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Dynamic Impact of Electrode Materials on Interface of Single-Crystalline Methylammonium Lead Bromide Perovskite

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One of the current challenges in methylammonium lead halide ($MAPbX_3$) perovskite application research is understanding contact formation and interfacial phenomena for highly efficient and stable device performance. For semiconductors, development of contact formation is inseparable from device performance and stability. Single-crystalline $MAPbX_3$ has become of great interest for perovskite devices in photodetectors, light-emitting diodes, and more recently in high-energy radiation detection. Deeper research is required to understand interfacial interactions in single-crystalline $MAPbX_3$. This article focuses on the dynamic impact of electrode metal (Au and Cr) on methylammonium lead bromide ($MAPbBr_3$) single crystals. It is studied how charge transport properties of single crystal $MAPbBr_3$ can be tuned via electrode material selection at the metal/$MAPbBr_3$ interface to improve device performance with proper contact formation. The ability to create an ohmic-like or nonohmic contact by switching the electrode metal from Cr to Au, respectively, is demonstrated. It is observed that the interfacial charge transfer resistance (recombination resistance) of the Cr/$MAPbBr_3$ interface is $1.79 \times 10^9\ \Omega$, compared to $1.32 \times 10^7\ \Omega$ for the Au/$MAPbBr_3$. Cr contacts can reduce hysteretic behavior by reducing interfacial recombination and interfacial polarization. These studies provide insight to metal/$MAPbX_3$ interfacial interactions toward device engineering for hole transport layer-free $MAPbX_3$ device structures.

1. Introduction

Methylammonium lead halide perovskites ($MAPbX_3$) have been vastly studied for a variety of applications including photovoltaics,[1–6] light-emitting diodes,[7] photodetectors,[8] lasing,[9] and recently for high energy radiation sensors.[10–12] These materials have gained much interest due to their tunable optical bandgaps, desirable electronic properties, and low cost.[13–15] Single-crystalline methylammonium lead halide has gained a lot of attention due to its low trap state density, higher mobility, and long carrier diffusion length in comparison to thin film $MAPbX_3$.[16,17] In addition, single crystalline $MAPbBr_3$ is grown using simple, low-temperature, solution-based crystal growth techniques, with the growth process only taking hours to days to complete. For device fabrication, $MAPbBr_3$ single crystals are easily processed as they grow cubic in shape with parallel sides without the need of complicated control methods during the growth.[18–20] These factors have made single crystalline $MAPbBr_3$ a promising material for use in the new generation of low cost electronic devices.

One large focus is on interface engineering via the choice of electrode materials, interfacial layers between the material and electrode and post growth processing before electrode deposition. Many interfacial layers and electrodes have been investigated in thin films for a variety of active materials.[21–25] Extensive research has been concluded on the impact of electrode material selection and interfacial engineering on the electronic properties of $MAPbX_3$ thin film devices.[26–29] The typical device structure for $MAPbX_3$ consists of an anode, an electron transport layer (ETL), the perovskite active layer, a hole transport layer (HTL), and a high work function cathode (e.g., silver, gold, chromium, etc.).[30] Although this device structure is commonly used for applications, a well-known alternative uses a HTL-free device structure. These HTL-free devices have been shown to reach high photovoltaic efficiencies,[31,32] and have been used for the newly discovered application in high-energy radiation detection.[11,12] Without the use of HTLs, it is important to understand how different metal electrodes can

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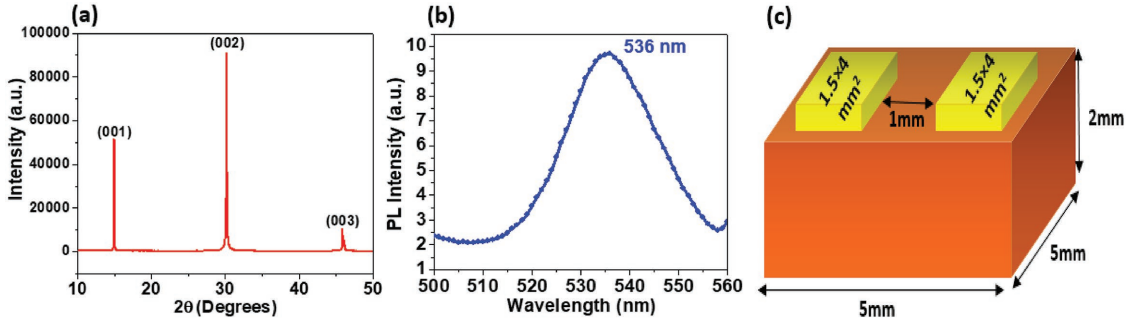

Figure 1. a) XRD for single crystal $MAPbBr_3$. b) PL for $MAPbBr_3$ single crystal (photoexcitation = 405 nm CW laser). c) Schematic of device architecture used throughout manuscript.

interact with $MAPbX_3$ through interfacial interactions. It has been found that the dominant process for charge transport properties and recombination in $MAPbX_3$ is fast vacancy-mediated anion diffusion.[33] Recently, there has been a great focus on research to understand how different interfaces between $MAPbX_3$ and single-material contact electrodes affect the charge transport and recombination properties via ion migration.[34,35] There has been a great effort to study interfacial effects in thin film $MAPbX_3$ devices. There is a need for research in contact engineering and interfacial phenomena for single crystalline $MAPbX_3$ devices with advantages such as the absence of grain boundaries and noncrystalline domains, toward deeper understanding of intrinsic properties and improvement of device performance for $MAPbX_3$-based devices and materials.

Here, we investigate the effects of the interface between single-metal electrodes and single-crystalline $MAPbBr_3$. The metal/semiconductor interfacial interactions are important to understand as HTL-free devices have been gaining attention due to exceptional hole transport properties of $MAPbX_3$. Because of this, device production is simplified for HTL-free devices without the necessity of deposition of interfacial layers between the active layer and electrode.[12,36] Ag and Au are among the most commonly used electrodes in $MAPbX_3$ optoelectronic devices, and in recent literature, Cr electrodes have been used in high-energy radiation detectors based on single-crystalline $MAPbBr_3$.[11,12,30] Silver was not chosen due to known chemical reactions that occur when in direct contact with $MAPbX_3$.[37] Therefore, Cr and Au were chosen as model systems to study the impact of metal electrode selection on the electronic properties of single crystalline $MAPbBr_3$ via interfacial interactions in dark condition, to exclude complexity of photoexcitation during analysis. Also, applications, such as high-energy radiation detection, perform under dark conditions. Cr and Au have work functions of 4.5 and 5.1–5.47 eV, respectively.[38] Cr has been used for single-crystalline $MAPbBr_3$ device fabrication, which resulted in linear IV response.[12] On the other hand, Au has been shown to create interfacial polarization with different materials including organic semiconductors and methylammonium lead halide perovskites.[25,39] Here, an in-plane electrode design is used to reduce bulk interactions and directly study the interface between $MAPbBr_3$ single crystals and the singular metal contact. Also, by using the capacitor-like metal/$MAPbBr_3$/metal device structure on one surface, we avoid any potential affects from ETLs in a typical device design. This allows us to explore direct correlation between charge transport properties and the interfacial phenomena in the $MAPbBr_3$/metal contact for development of single-crystalline HTL-free device structures.

2. Results and Discussion

Figure 1a,b shows the X-ray diffraction (XRD) and photoluminescence data for single-crystalline $MAPbBr_3$ studied in this manuscript. The peaks observed in XRD data, and the PL peak at 536 nm agrees with previous literature and shows that the material studied is single crystalline $MAPbBr_3$.[40] Figure 1b shows the device structure schematic for the samples.

For this device structure, a 5 × 5 × 2 mm crystal was studied as-grown. Two contacts were thermally evaporated on the flat surface of the crystal with a thickness of 60 nm. These contacts were deposited with a width of 1 mm between them with electrode sizes of 4 mm long and 1.5 mm wide. **Figure 2**a,b shows the IV curves for $MAPbBr_3$ with Cr and Au contacts, respectively. As shown in Figure 2a, an ohmic-like

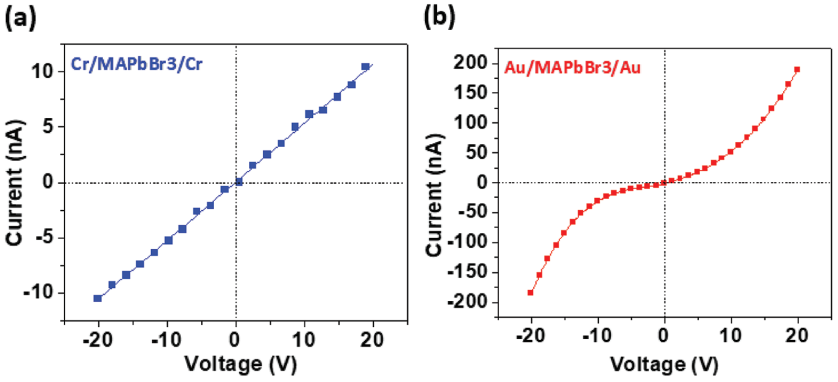

Figure 2. IV curves for single-crystalline $MAPbBr_3$ with lateral design device architecture using a) Cr/Cr contacts and b) Au/Au contacts.

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contact is formed at the Cr/$MAPbBr_3$ interface, while a nonohmic contact is formed at the Au/$MAPbBr_3$ interface. Here, we define the ohmic-like behavior as linear dark current in both positive and negative bias directions. The nonohmic behavior is defined as dark current with exponential behavior. The nonohmic IV characteristic in $MAPbBr_3$ single crystals with Au contacts was also reported in earlier studies.[41] These characteristics are reproducible across several single-crystal devices with Cr and Au contact materials in ambient conditions. Reproducibility is shown in Figures S1 and S2 of the Supporting Information confirming contact phenomena with Cr and Au, ruling out the effects of environment on contact formation. Generally, the interface between a semiconductor and electrode materials facilitates the charge transport by providing an energetic offset between the conduction bands.

The bandgap of $MAPbBr_3$ is ≈2.3 eV, with the conduction band at −4.2 eV and the valence band at −6.5 eV.[42] The ohmic-like and nonohmic contact formation in Cr and Au, respectively, can be attributed to the band alignment between the Cr and Au contacts and $MAPbBr_3$. There is a small potential barrier of around 0.3 eV between the conduction band of $MAPbBr_3$ and the work function of Cr. Therefore, the small band offset does not limit the current flow through the device, shown by the linear increase and decrease with voltage. On the other hand, considering the work function of Au (5.1–5.47 eV), the potential barrier between $MAPbBr_3$ and Au is much larger (≈0.9–1.37 eV). Because of this larger potential barrier, charge carriers require more energy to move from the single crystal to Au and the nonohmic behavior is observed with Au contacts.[43] In addition, Au has previously been shown to migrate through $MAPbX_3$ materials under illumination and heat, severely affecting the performance of devices under working conditions.[28,44,45] Therefore, control over contact formation and contact phenomena is inseparable from improvement in perovskite device performance and stability. For $MAPbX_3$ devices, ohmic and nonohmic contacts can be useful for many diverse applications. Ohmic contacts are desirable for applications such as LEDs (better charge injection) and high-energy radiation detection (better charge collection), whereas nonohmic contacts are more common for applications such as solar cells. Next, we explored the hysteretic behavior of each device with current–voltage (IV) characteristics, as shown in **Figure 3**.

Figure 3a shows IV sweeps at different scan rates for the Cr sample. A small amount of hysteresis is observed the device with Cr contacts. Figure 3b,c shows hysteresis behavior with the Au contacts. The hysteresis increases significantly in the sample with Au contacts as scan rate decreases from 50 to 0.5 V s^{-1}, and it is also time dependent through multiple scans, as seen in Figure 3b,c, respectively. (Figure S3, Supporting Information, shows multiple IV hysteresis characteristics as a function of scan rate and multiple scan cycles for Au devices.) Minimizing hysteresis has been one of the main challenges toward the development of device performance and stability.[33–37,43,46,47] Comparing Cr and Au contacts via IV sweeps, we observe the Cr/$MAPbBr_3$ interface greatly reduces hysteresis effects as compared to the Au/$MAPbBr_3$ interface. Several factors can influence hysteresis behavior in $MAPbX_3$ devices including vacancy-mediated halide migration, interfacial recombination, band alignment, and trap states.[33,48,49] Ion migration in $MAPbX_3$ is discussed in detail by Frost and Walsh, showing that halide migration in $MAPbX_3$ active layers and interfacial recombination are the dominant mechanisms attributing to device hysteresis.[33] Therefore, we propose that greater hysteresis observed with Au contacts is attributed to two factors: 1) interfacial recombination is greater at the Au/$MAPbBr_3$ interface and 2) the Au/$MAPbBr_3$ interface increases the contribution of Br^- migration to the electronic properties of $MAPbBr_3$ and an increase in interfacial recombination compared to the Cr/$MAPbBr_3$ interface. To gain a deeper understanding about the dynamic processes at the interface between $MAPbBr_3$ single crystals and metal electrodes, impedance spectroscopy was performed for samples with different metal electrodes (**Figure 4**a,b).

At the mid to high frequency, we are comparing the bulk properties away from the interface. At low frequency, we examine the interface directly.[50] Low frequency impedance measured on Au electrodes exhibits inductive behavior (Figure 4b). This behavior was observed previously for single-crystalline $MAPbBr_3$ and is believed to arise from electronic carrier injection from Au into the perovskite material.[41] Inductive effects at $MAPbBr_3$/Au interfaces will be more thoroughly explored in follow-up experiments. In this study, we used a simplified RC circuit, shown in the inset of Figure 4a, to make a direct comparison between the two samples at the metal/$MAPbBr_3$ interfaces. This equivalent circuit is adapted from previous literature on interfacial properties of $MAPbI_3$ thin films studied

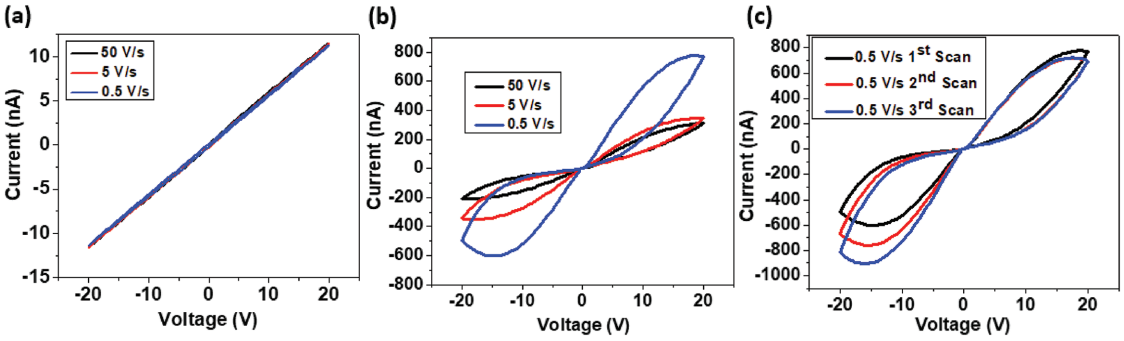

Figure 3. IV sweeps for single crystalline $MAPbBr_3$ with in-plane design device architecture using a) Cr/Cr contacts (different scan speeds), b) Au/Au contacts (different scan speeds) and c) Au/Au Contacts (same scan speeds, multiple scans).

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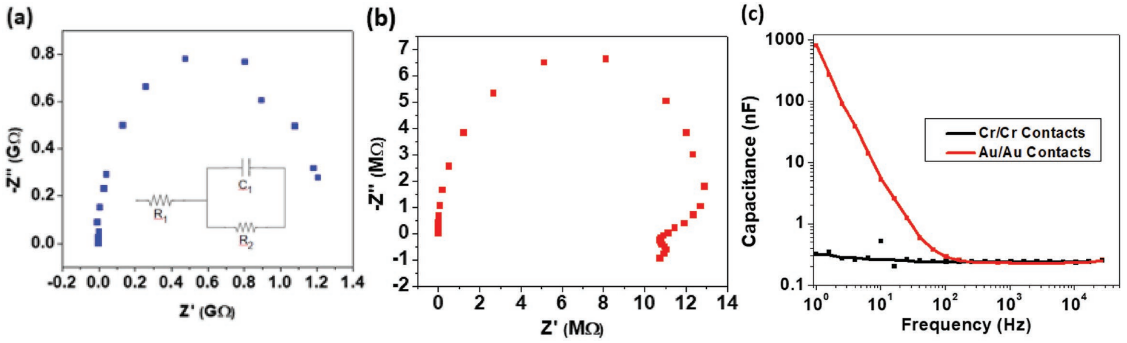

Figure 4. Nyquist plots and modeling for single crystalline $MAPbBr_3$ with lateral design device architecture using a) Cr/Cr contacts and b) Au/Au contacts. c) Capacitance as a function of frequency from impedance spectroscopy.

via impedance spectroscopy.[51] Using this equivalent circuit as representation of the device, we correspond the R1 to the series resistance, as the resistance of Cr and Au are negligible to the overall device. Then, we correspond the R2 and C1 values to the metal/$MAPbBr_3$ interface. In this perspective, the RC equivalent circuit focuses on differences in the resistive and capacitive nature of the Cr/$MAPbBr_3$ and Au/$MAPbBr_3$ interfaces.

Table 1 shows the data calculated by the modeling of the two impedance spectra from Figure 4. In both samples, R1 is shown to be very close to zero, confirming good contact formation with negligible resistance. Above frequencies of 100 Hz, the capacitance for both devices is shown to be very similar, on the order of 10^{-10} F. However, the large difference is shown in resistance at R2. The Cr device has a larger resistance of 1.79 GΩ, while the Au device shows a decrease in resistance to 13.2 MΩ. Based on these values, the RC-time constants (τ) were calculated for each device. The Cr device shows a relatively long RC time constant, $\tau = 0.5997$ s. This shows the behavior of the Cr sample is heavily influenced by the high recombination resistance of the Cr/$MAPbBr_3$. This is also observed by the dark IV characteristics, where the Cr contacts show a much lower dark current, as compared to the Au contacts (Figure 2). The Au has a much shorter RC time constant, $\tau = 0.003$ s, possibly due to the capacitive nature of the Au/$MAPbBr_3$ at low frequency, observed in Figure 4c. Increasing the interfacial charge transfer resistance has been shown to reduce the electron recombination at the interface to enhance photovoltaic properties of $MAPbI_3$ devices.[51] Similarly, we expect that increasing the resistance of the $MAPbBr_3$/metal interface by using Cr over Au will reduce recombination at the interface and improve charge collection efficiencies in $MAPbX_3$ devices.

Figure 4c shows the frequency-dependent capacitance of both samples with Cr and Au contacts. Below 100 Hz, the interfaces of the samples are examined. The sample with Cr contacts shows a very small increase in capacitance from 0.23 to 0.31 nF from 100 to 1 Hz. Meanwhile, the sample with Au contacts has a capacitance increase by a factor of $\approx 10^4$ going from 0.3 nF to 1 μF from 100 to 1 Hz. The increase in capacitance at low frequency gives direct evidence to compare interfacial polarization at the metal/$MAPbBr_3$ interfaces. The larger capacitance at $MAPbBr_3$/Au interface clearly shows larger accumulation of both electronic and ionic charge carriers which was previously revealed as the mechanism behind IV hysteresis in perovskite devices.[33]

Table 1. Resistance and capacitance values for two different samples based on a simple RC circuit modeling.

Equivalent circuit component	Cr/Cr contacts	Au/Au contacts
R_1 [Ohms]	1.08×10^{-8}	8.42×10^{-10}
R_2 [Ohms]	1.79×10^{9}	1.32×10^{7}
C_1 [F]	3.5×10^{-10}	2.28×10^{-10}
RC-τ [s]	0.5997	0.0030

We further explored transient response of the crystals using Cr and Au contacts (**Figure 5**). In this measurement, a voltage power supply was connected to one contact to bias the device with a 5 V pulse. Then, the device was connected to an oscilloscope to measure the output of the device (Figure 5a inset). In this way, we examine the voltage drop across the device, where a smaller output voltage indicates a larger voltage drop. Because the only variability in the experiments is the electrode metal, (Cr or Au) the differences examined between the two devices directly corresponds to the changes in interactions at the metal/$MAPbBr_3$ interface. In Figure 5a,b, the voltage pulse is represented by the left-hand y-axis, and the voltage output of the device is represented by the right-hand y-axis. First, the rise and decay times of the voltage through the device were compared when the square pulse was turned on and off. In Figure 5a, the Cr device shows a rapid response with the square pulse turning on and off. On the other hand, in Figure 5b, the Au device shows a slow rise and decay time in response to the square pulse. Jacobs et al. describe how slow transient responses are attributed to both charge trapping and ion migration in $MAPbI_3$.[52] When the voltage is turned off for the Cr samples, the output voltage becomes negative for a short time before relaxing back to zero. This relaxation was fitted as shown in Figure 5d, showing a relaxation time of 0.48 ms. The fast response to voltage pulses can be attributed to the Cr/$MAPbBr_3$ interface, where the contact is ohmic and current flow is only restricted by the internal resistance of the device. In addition, because the interfacial resistance is high at the Cr/$MAPbBr_3$ interface, interfacial recombination and charge trapping are minimized, resulting in the fast response to the voltage pulses. In Figure 5c, it is shown that this process enhances the resistance of the device as the output voltage decreases over time under repetitive pulsed voltages. It is a small decrease of ≈5 mV after 2000 s of repetitive voltage pulsing. The increase in resistance can be explained by charge/ion redistribution to an equilibrium state after 1000 s of voltage pulses, as the change in output voltage

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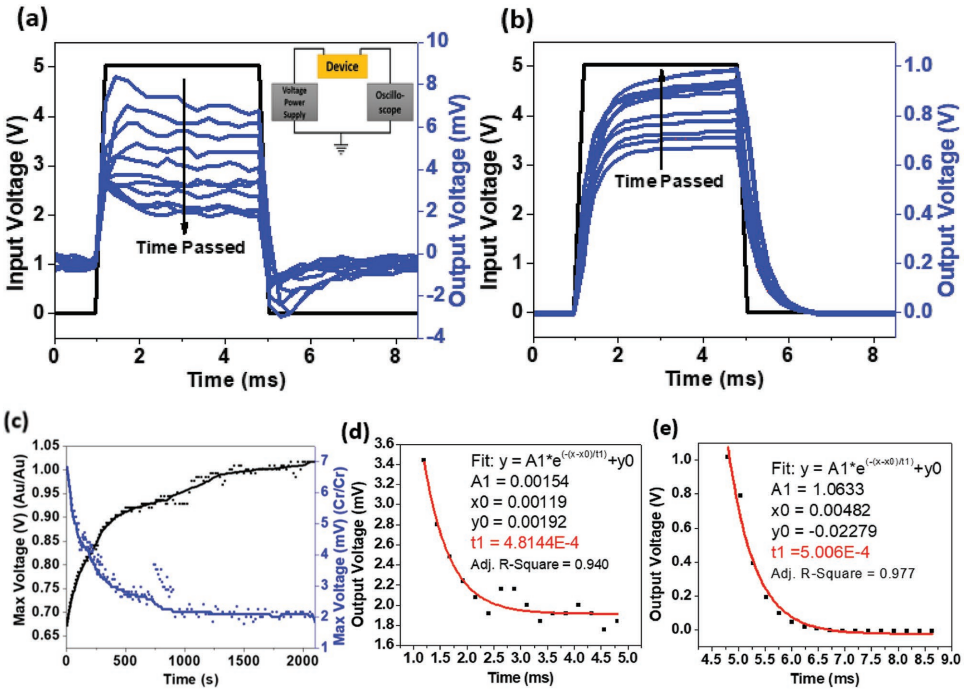

Figure 5. Pulse voltage measurements with a 5 V square wave input for $MAPbBr_3$ a) Cr/Cr (0-1500 s) and b) Au/Au contacts (0–1500 s). c) Max voltage as a function of time for both samples. (Max voltage is considered the voltage at 5 ms). d,e) Decay relaxation fit for Cr/Cr and Au/Au contacts, respectively. (5c time scale shows the evolution of pulse voltages for full 35 min. 5a and 5b only show 1500 s to avoid cluttering of data.)

significantly decreases at this time. The Au contacts show a slow rise (when voltage is on) and decay (when voltage is off). The relaxation here was fitted in Figure 5e, giving a relaxation time of 0.5 ms. In Figure 5c, it is shown that this process reduces the resistance of the device as the output voltage increases over time under repetitive pulse voltages. Because the Au/$MAPbBr_3$ interface was shown to have relatively high interfacial polarization from impedance spectroscopy, we attribute the slow response of the output voltage to high interfacial polarization due to ion migration and accumulation of charges, as well as charge trapping at the interface. Therefore, we show that ion migration has a large impact on the electronic properties at the Au/$MAPbBr_3$ interface, as compared to the Cr/$MAPbBr_3$ interface. It was previously shown in thin film perovskite devices with inverted structure that the accumulation of ions at the perovskite/electrode interface results in decreasing the Schottky barrier height, resulting in an increase in interfacial resistance.[53] This is confirmed here by the increasing output voltage through time (Figure 5c). Combining this information with hysteresis, we show that hysteresis can be heavily reduced from $MAPbX_3$-based optoelectronic devices by choosing the correct electrode metal with high interfacial resistance and low interfacial polarization. The ion redistribution and ion migration proposed here are both shown to be reversible processes as the hysteresis in both contact schemes returns to the starting current position at zero voltage. Further investigation would be required for full device structures under working conditions to better understand the reversibility or irreversibility of these processes at metal/$MAPbBr_3$ interfaces, as well as further exploration on the influence of chemical interactions on *I–V* and impedance characteristics at the metal–$MAPbX_3$ interfaces.

3. Conclusions

In summary, we investigated the effects of electrode material on the electrical properties in single crystalline $MAPbBr_3$ devices. We found that Cr contacts greatly reduce the amount of hysteresis at the metal/$MAPbBr_3$ interface, compared to the large hysteresis observed with Au contacts. We show that the Cr contacts increase the interfacial resistance of the sample by over a factor of 100, which reduces the amount of interfacial recombination. We also observed that Au creates a much stronger interfacial polarization leading to increased contribution of Br^- migration to the electronic properties. In this way, we conclude that using Cr electrodes at a direct Cr/$MAPbBr_3$ interface greatly reduces hysteresis effects via combined properties of weak interfacial polarization to reduce contribution of ion migration in $MAPbBr_3$ and stronger interfacial resistance to reduce interfacial recombination forming an ohmic-like contact, whereas the Au/$MAPbBr_3$ interface forms a nonohmic contact. We believe that this work provides deeper understanding on ion migration and recombination properties at interfaces and emphasizes the significant impact of interfacial effects at metal/$MAPbBr_3$ on electronic properties of $MAPbBr_3$ single crystalline devices toward future application research using HTL-free device structures.

4. Experimental Section

Growth of $MAPbBr_3$: Methylammonium lead bromide ($MAPbBr_3$) single crystals were grown using the inverse temperature method.[18] To prepare the growth solution, methylammonium bromide and

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lead(II) bromide ($PbBr_2$) were mixed with equimolar amounts in *N,N*-dimethylformamide (DMF). The solution was constantly stirred for 1 h at room temperature. After completely dissolving, the solution was then filtered using a 0.2 µm PTFE filter. After filtration, the solution was divided among vials in 2 mL quantities. The vials were then placed in an oil bath and grown overnight at 75 °C. After growth, the samples were washed using dichloromethane (DCM). All materials used in the growth process were purchased from Sigma-Aldrich and used as received.

Electrode Deposition: Thermal evaporation was used for electrode deposition of Cr and Au. Each electrode was deposited with a thickness of 60 nm using an in-plane electrode design. The in-plane electrode design was chosen to reduce the bulk interactions to study the surface and interfacial effects.

Measurements Techniques: All measurements completed in this article were completed in dark ambient conditions due to the high sensitivity of $MAPbBr_3$ to light. Tungsten probe tips were used to contact the sample for each measurement in the same arrangement. IV measurements were completed using a Keithley voltage source from −20 to +20 V at a rate of 0.25 V s^{-1}. IV hysteresis was measured with the same setup, using LabView to program the scans at different scan rates (0.5, 5, and 50 V s^{-1}). For both samples, impedance was measured from 1 Hz to 6 MHz with 0 VDC bias and 1 VAC excitation. EIS Spectrum Analyser was used for modeling impedance traces. The pulsed voltage measurements were conducted using a pulse generator to create a 130 Hz 5 V square wave. The bias was applied to one contact, then the output voltage was measured at the other contact with an oscilloscope in series to study the voltage drop across the device. Photoluminescence was completed with a 405 nm CW laser for photoexcitation.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

hysteresis, impedance spectroscopy, interfaces, ion migration, metal electrodes, methylammonium lead halide perovskites

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